

$$I = \int_0^{\pi/2} \frac{\cos x - \sin x \cos x}{1 + \sin x} dx$$

$$= \int_0^{\pi/2} \frac{\cos x}{1 + \sin x} dx - \int_0^{\pi/2} \frac{\sin x \cos x (1 + \sin x)}{(1 + \sin x)(1 + \sin x)} dx$$

$$= (\ln |1 + \sin x|)_0^{\pi/2} - \int_0^{\pi/2} \frac{\sin x \cos x}{1 + \sin x} dx$$

$$= \ln 2 - \int_0^{\pi/2} \frac{\sin x}{\cos x} - \frac{\sin^2 x}{\cos x} dx$$

$$= \ln 2 + \int_0^{\pi/2} \frac{-\sin x}{\cos x} dx + \int_0^{\pi/2} \left(\frac{1 - \cos^2 x}{\cos x} \right) dx$$

$$= \ln 2 + (\ln |\cos x|)_0^{\pi/2} + \int_0^{\pi/2} \cos x dx + \int_0^{\pi/2} \frac{1}{\cos x} dx$$

$$= \ln 2 + \lim_{x \rightarrow \pi/2} \ln |\cos x| - [\sin x]_0^{\pi/2} + \int_0^{\pi/2} \frac{\cos x}{(1 + \sin x)(1 + \sin x)} dx$$

$$= \ln 2 - 1 + \lim_{x \rightarrow \pi/2} \ln |\cos x| + \frac{1}{2} \int_0^{\pi/2} \frac{\cos x}{1 + \sin x} + \frac{\cos x}{1 + \sin x} dx$$

$$= \ln 2 - 1 + \lim_{x \rightarrow \pi/2} \ln |\cos x| + \frac{1}{2} [-\ln |1 + \sin x| + \ln |1 + \sin x|]_0^{\pi/2}$$

$$= \ln 2 - 1 + \lim_{x \rightarrow \pi/2} \ln |\cos x| + \frac{1}{2} \left(\ln |1 + \sin x| \right)_0^{\pi/2}$$

$$= \ln 2 - 1 + \lim_{x \rightarrow \pi/2} \left(\ln |\cos x| - \frac{1}{2} \ln |1 + \sin x| \right) + \frac{1}{2} (\ln 2)$$

$$= \frac{3}{2} \ln 2 - 1 + \lim_{x \rightarrow \pi/2} \ln \left| \frac{\cos x}{\sqrt{1 + \sin x}} \right|$$

$$+ \lim_{x \rightarrow 0} \ln \left| \frac{\cos x}{\sqrt{1 + \sin x}} \right|$$

$$I = \lim_{y \rightarrow 0} \ln \left| \frac{\cos(y + \pi/2)}{\sqrt{1 - \sin(y + \pi/2)}} \right|$$

$$I = \lim_{n \rightarrow \pi/2} \frac{\cos n}{\sqrt{1 - \sin n}} = \lim_{n \rightarrow \pi/2} \sqrt{\frac{\cos^2 n \times 1 + \sin n}{(1 - \sin n) 1 + \sin n}}$$

$$= \lim_{n \rightarrow \pi/2} \sqrt{1 + \sin n} = \sqrt{2}$$

$$I_2 = \lim_{n \rightarrow \pi/2^+}$$

$$I = \frac{3}{2} \ln 2 - 1 + \lim_{n \rightarrow \pi/2} \frac{\cos n}{\sqrt{1 - \sin n}}$$

$$+ \lim_{n \rightarrow \pi/2} \sqrt{\frac{\cos^2 n \times 1 + \sin n}{(1 - \sin n) 1 + \sin n}}$$

$$+ \lim_{n \rightarrow \pi/2} \sqrt{1 + \sin n}$$

$$I = \frac{3}{2} \ln 2 - 1 + \ln \sqrt{2} = 2 \ln 2 - 1$$

$$I = \int_0^{\pi/2} \frac{\cos x - \sin x \cos x}{1 + \sin x} dx$$

$$= \int_0^{\pi/2} \frac{\cos x (1 - \sin x)}{1 + \sin x} dx$$

$$= \int_0^{\pi/2} \cos x \left(-1 + \frac{2}{1 + \sin x} \right) dx$$

$$= \int_0^{\pi/2} \left(-\cos x + \frac{2\cos x}{1 + \sin x} \right) dx$$

$$= \left[-\sin x + 2 \ln |1 + \sin x| \right]_0^{\pi/2}$$

$$I = -1 + 2 \ln 2 = \ln \left(\frac{4}{e} \right)$$

Deduction $J_2 = \int_0^{\pi/2} \frac{\sin x \cos x}{1 + \sin x} dx = \int_0^{\pi/2} \frac{\cos x (1 + \sin x) + \cos x}{1 + \sin x} dx$

$$= \cancel{1} - I + \left[\ln |1 + \sin x| \right]_0^{\pi/2}$$

$$= 1 - 2 \ln 2 + \ln 2$$

$$= 1 - \ln 2$$

$$= \ln \left(\frac{e}{2} \right)$$